

Table 1. Spanish-input, English-output TruthfulQA transfer results using English-calibrated controllers. Full-protocol rows report mean $\pm$ standard error across five decoding repetitions; refreshed Llama-3-8B H_∞ , GPT-2 XL, OLMo-2-32B, and compact Qwen-2.5-32B rows report five-group delete-one-group question-jackknife standard errors.

	Method	True (%) $\uparrow$	Informative (%) $\uparrow$	T $\times$ I (%) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Gemma-2-2B	Original	51.41 $\pm$ 0.90	86.44 $\pm$ 0.58	44.44 $\pm$ 0.83	1.32 $\pm$ 0.02	1.17 $\pm$ 0.00
	ITI	53.56 $\pm$ 0.47	88.59 $\pm$ 0.45	47.45 $\pm$ 0.48	1.42 $\pm$ 0.01	1.21 $\pm$ 0.01
	ActAdd	63.57 $\pm$ 0.39	67.15 $\pm$ 0.34	42.69 $\pm$ 0.34	0.99 $\pm$ 0.01	0.88 $\pm$ 0.01
	Mean-AcT	48.13 $\pm$ 0.52	90.40 $\pm$ 0.46	43.51 $\pm$ 0.52	1.42 $\pm$ 0.01	<u>1.23 $\pm$ 0.01</u>
	Linear-AcT	47.83 $\pm$ 0.78	90.60 $\pm$ 0.78	43.34 $\pm$ 0.80	<u>1.44 $\pm$ 0.01</u>	1.23 $\pm$ 0.01
	PID-AcT	48.79 $\pm$ 0.61	<u>90.89 $\pm$ 0.49</u>	44.35 $\pm$ 0.60	1.43 $\pm$ 0.01	1.22 $\pm$ 0.01
	ODESteer	57.01 $\pm$ 0.20	86.27 $\pm$ 0.28	49.18 $\pm$ 0.24	1.32 $\pm$ 0.01	1.14 $\pm$ 0.01
	S-PID	55.15 $\pm$ 0.37	86.41 $\pm$ 0.34	47.66 $\pm$ 0.37	1.29 $\pm$ 0.01	1.12 $\pm$ 0.00
	A-LQR	73.81 $\pm$ 0.71	76.55 $\pm$ 0.65	<u>56.50 $\pm$ 0.72</u>	1.12 $\pm$ 0.01	1.06 $\pm$ 0.00
	H_∞ (ours)	77.45 $\pm$ 0.52	64.16 $\pm$ 0.85	49.70 $\pm$ 0.74	1.04 $\pm$ 0.01	1.02 $\pm$ 0.01
Llama-3-8B	Original	42.86 $\pm$ 0.49	94.52 $\pm$ 0.24	40.51 $\pm$ 0.47	1.59 $\pm$ 0.01	1.39 $\pm$ 0.01
	ITI	55.47 $\pm$ 0.88	94.37 $\pm$ 0.20	52.35 $\pm$ 0.84	1.61 $\pm$ 0.02	1.39 $\pm$ 0.01
	ActAdd	43.94 $\pm$ 0.53	93.78 $\pm$ 0.29	41.21 $\pm$ 0.51	1.57 $\pm$ 0.01	1.35 $\pm$ 0.01
	Mean-AcT	46.61 $\pm$ 0.69	94.83 $\pm$ 0.49	44.20 $\pm$ 0.69	1.61 $\pm$ 0.01	1.41 $\pm$ 0.01
	Linear-AcT	47.15 $\pm$ 0.41	94.96 $\pm$ 0.39	44.77 $\pm$ 0.43	1.61 $\pm$ 0.01	<u>1.42 $\pm$ 0.02</u>
	PID-AcT	47.12 $\pm$ 0.63	<u>95.08 $\pm$ 0.51</u>	44.80 $\pm$ 0.64	<u>1.62 $\pm$ 0.01</u>	1.41 $\pm$ 0.01
	ODESteer	<u>95.67 $\pm$ 0.15</u>	11.21 $\pm$ 0.45	10.73 $\pm$ 0.43	0.00 $\pm$ 0.00	0.02 $\pm$ 0.00
	S-PID	52.09 $\pm$ 0.56	94.37 $\pm$ 0.22	49.16 $\pm$ 0.54	1.59 $\pm$ 0.01	1.33 $\pm$ 0.01
	A-LQR	55.57 $\pm$ 0.71	95.03 $\pm$ 0.25	52.81 $\pm$ 0.69	1.56 $\pm$ 0.01	1.27 $\pm$ 0.01
	H_∞ (ours)	68.54 $\pm$ 0.95	80.66 $\pm$ 0.81	55.29 $\pm$ 0.35	1.24 $\pm$ 0.02	1.05 $\pm$ 0.01
Qwen-2.5-14B	Original	68.13 $\pm$ 0.28	97.31 $\pm$ 0.13	66.29 $\pm$ 0.28	1.82 $\pm$ 0.00	1.51 $\pm$ 0.01
	ITI	67.91 $\pm$ 0.33	97.36 $\pm$ 0.07	66.11 $\pm$ 0.33	1.82 $\pm$ 0.00	1.53 $\pm$ 0.01
	ActAdd	72.09 $\pm$ 0.90	89.62 $\pm$ 1.54	64.61 $\pm$ 1.37	1.63 $\pm$ 0.04	1.37 $\pm$ 0.04
	Mean-AcT	68.27 $\pm$ 0.26	97.48 $\pm$ 0.08	66.55 $\pm$ 0.26	1.82 $\pm$ 0.01	<u>1.55 $\pm$ 0.01</u>
	Linear-AcT	68.25 $\pm$ 0.21	<u>97.55 $\pm$ 0.19</u>	66.58 $\pm$ 0.24	<u>1.83 $\pm$ 0.01</u>	1.55 $\pm$ 0.01
	PID-AcT	68.27 $\pm$ 0.22	97.28 $\pm$ 0.20	66.42 $\pm$ 0.25	1.82 $\pm$ 0.01	1.55 $\pm$ 0.02
	ODESteer	74.88 $\pm$ 0.46	92.83 $\pm$ 0.26	69.51 $\pm$ 0.47	1.67 $\pm$ 0.01	1.44 $\pm$ 0.01
	S-PID	82.82 $\pm$ 0.28	93.56 $\pm$ 0.47	77.48 $\pm$ 0.47	1.64 $\pm$ 0.00	1.43 $\pm$ 0.01
	A-LQR	85.24 $\pm$ 0.50	94.32 $\pm$ 0.27	80.40 $\pm$ 0.52	1.64 $\pm$ 0.01	1.41 $\pm$ 0.01
	H_∞ (ours)	88.74 $\pm$ 0.28	93.05 $\pm$ 0.48	82.57 $\pm$ 0.50	1.53 $\pm$ 0.01	1.29 $\pm$ 0.01
Qwen-2.5-32B	Original	72.86 $\pm$ 3.53	94.13 $\pm$ 0.45	68.59 $\pm$ 3.34	1.68 $\pm$ 0.04	1.42 $\pm$ 0.02
	S-PID	<u>86.06 $\pm$ 1.20</u>	89.73 $\pm$ 2.05	77.23 $\pm$ 2.07	1.57 $\pm$ 0.05	1.32 $\pm$ 0.03
	A-LQR	84.84 $\pm$ 2.13	<u>92.42 $\pm$ 1.10</u>	<u>78.41 $\pm$ 2.18</u>	<u>1.60 $\pm$ 0.04</u>	<u>1.39 $\pm$ 0.02</u>
	H_∞ (ours)	84.84 $\pm$ 0.85	89.24 $\pm$ 0.48	75.71 $\pm$ 0.86	1.45 $\pm$ 0.04	1.30 $\pm$ 0.03
OLMo-2-32B	Original	62.55 $\pm$ 2.00	97.55 $\pm$ 0.61	61.01 $\pm$ 1.81	1.76 $\pm$ 0.02	1.55 $\pm$ 0.01
	A-LQR	90.45 $\pm$ 1.22	91.43 $\pm$ 1.14	82.70 $\pm$ 1.14	1.41 $\pm$ 0.01	1.02 $\pm$ 0.01
	H_∞ (ours)	95.59 $\pm$ 0.71	95.72 $\pm$ 0.61	91.50 $\pm$ 1.07	1.53 $\pm$ 0.01	1.08 $\pm$ 0.01
GPT-2 XL	Original	66.10 $\pm$ 0.86	11.51 $\pm$ 0.77	7.60 $\pm$ 0.44	0.05 $\pm$ 0.01	0.15 $\pm$ 0.02
	ITI	60.10 $\pm$ 1.13	16.52 $\pm$ 1.65	9.93 $\pm$ 0.95	<u>0.10 $\pm$ 0.01</u>	<u>0.33 $\pm$ 0.01</u>
	ActAdd	65.61 $\pm$ 2.12	14.44 $\pm$ 1.04	9.48 $\pm$ 0.52	0.05 $\pm$ 0.01	0.24 $\pm$ 0.02
	Mean-AcT	69.65 $\pm$ 1.32	12.97 $\pm$ 1.64	9.04 $\pm$ 1.22	0.02 $\pm$ 0.00	0.07 $\pm$ 0.00
	Linear-AcT	61.93 $\pm$ 0.84	14.44 $\pm$ 1.18	8.95 $\pm$ 0.74	0.10 $\pm$ 0.01	0.26 $\pm$ 0.01
	PID-AcT	65.85 $\pm$ 3.25	13.95 $\pm$ 0.76	9.19 $\pm$ 0.65	0.07 $\pm$ 0.01	0.21 $\pm$ 0.01
	ODESteer	73.93 $\pm$ 1.61	12.48 $\pm$ 1.01	9.23 $\pm$ 0.81	0.07 $\pm$ 0.00	0.13 $\pm$ 0.01
	S-PID	67.20 $\pm$ 1.70	10.89 $\pm$ 1.39	7.32 $\pm$ 1.00	0.02 $\pm$ 0.00	0.07 $\pm$ 0.01
	A-LQR	74.30 $\pm$ 0.93	8.69 $\pm$ 0.80	6.46 $\pm$ 0.58	0.06 $\pm$ 0.01	0.11 $\pm$ 0.01
	H_∞ (ours)	84.21 $\pm$ 1.55	21.54 $\pm$ 1.68	18.14 $\pm$ 1.31	0.04 $\pm$ 0.01	0.06 $\pm$ 0.01